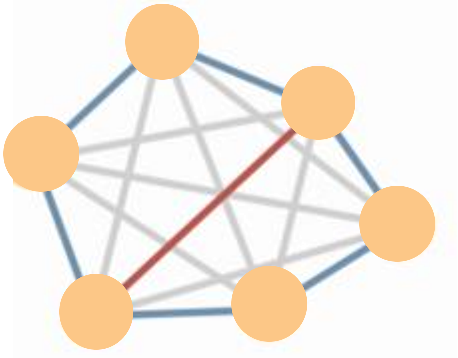
Bayesian Graphical Modeling

Welcome to the workshop!

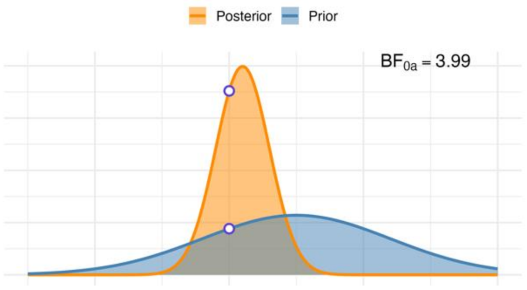
BGM workshop, Tübingen, 2025

m.marsman@uva.nl
bayesiangraphicalmodeling.com

Introduction to the workshop



Network Psychometrics

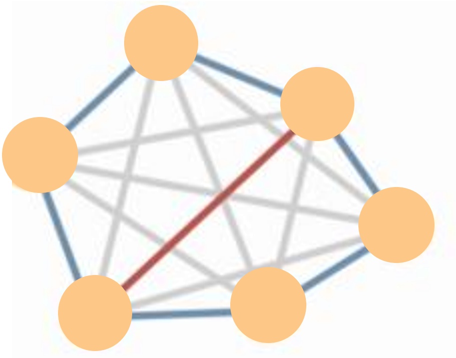


Bayesian Statistics

This Workshop

image: Flaticon.com

Introduction to the workshop

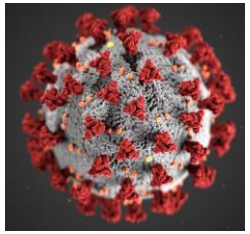


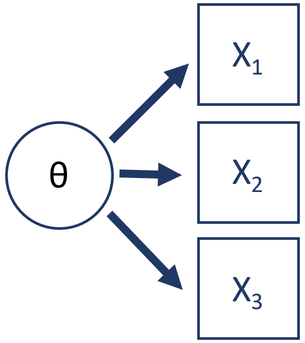
Network Psychometrics

image: Flaticon.com

Latent variable model vs network model

Disease analogy

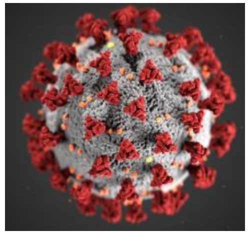




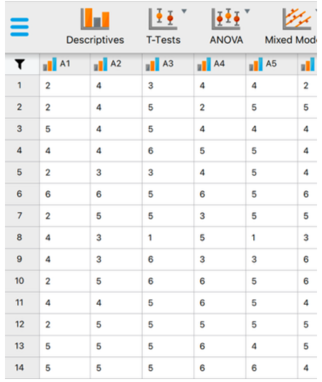
Latent variable model

Latent variable model vs network model

Disease analogy



Psychological data



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8	4	3	1	5	1	3
9	4	3	6	3	3	6
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11	4	4	5	6	5	4
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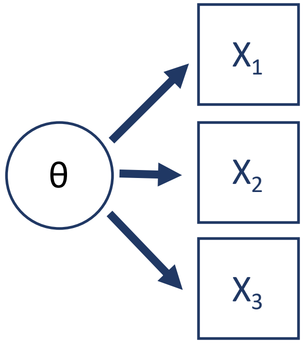
13	5	5	5	6	4	5
14	5	5	5	6	6	4

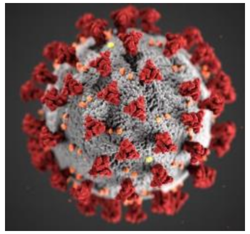


Latent variable model

No conclusive evidence for single cause

Latent variable model vs network model

Disease analogy



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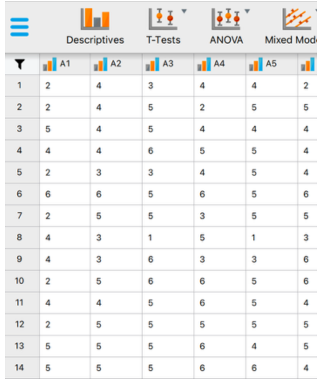

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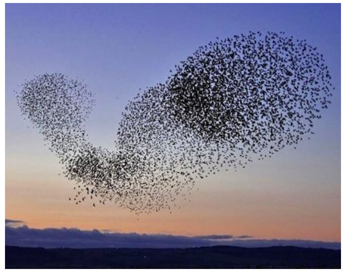
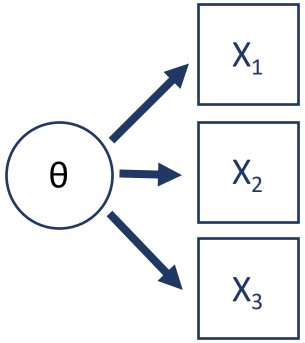
Flocking analogy





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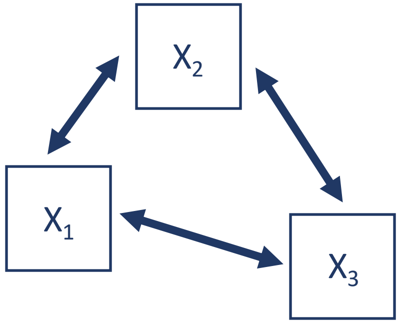
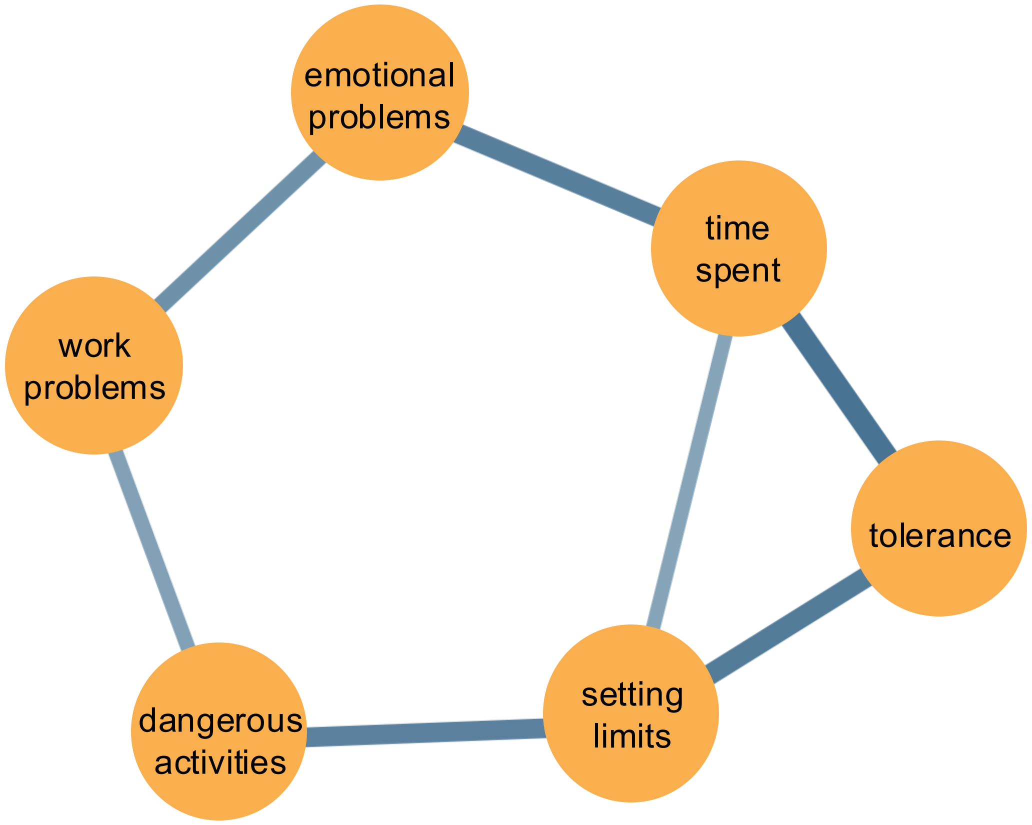


Network model

Psychometric network of AUD symptoms



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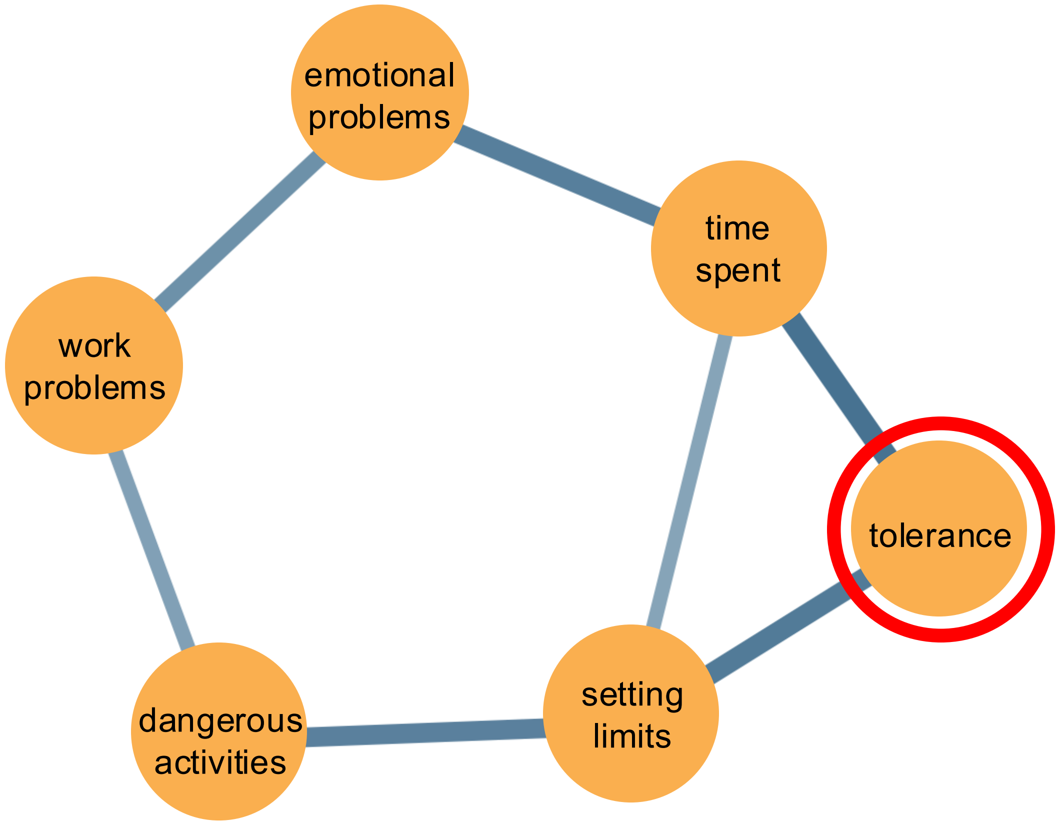
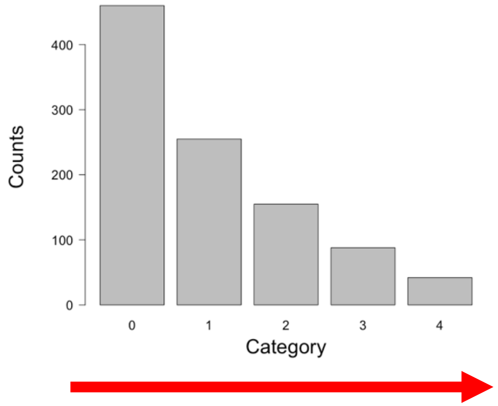
Psychometric network of AUD symptoms





symptom severity

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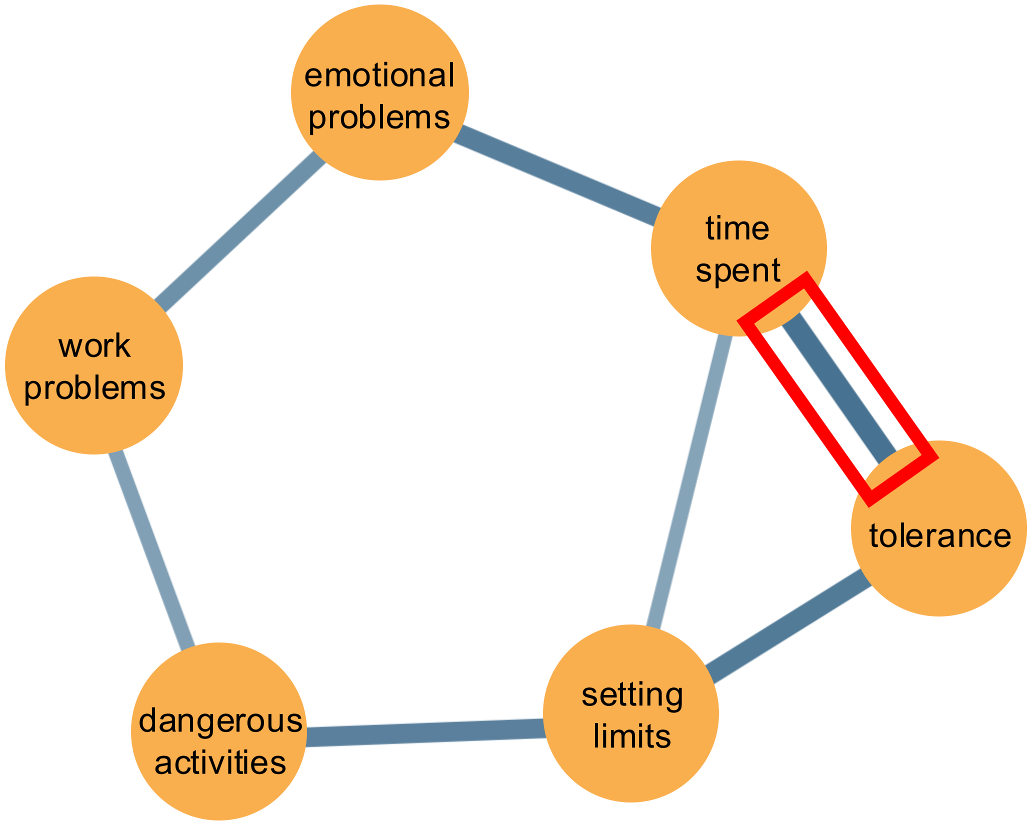

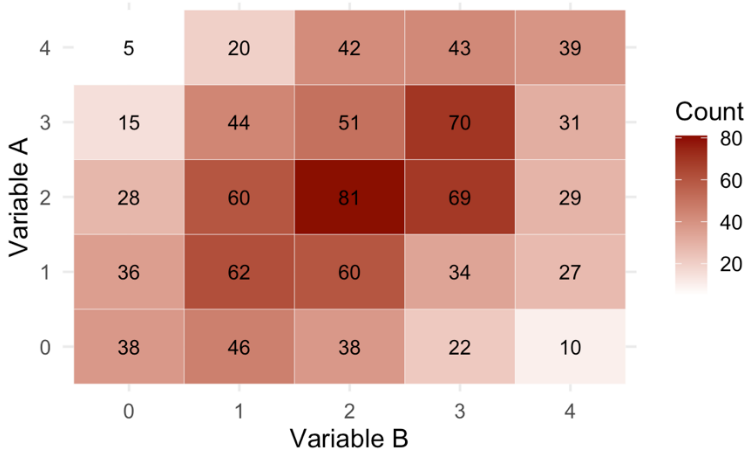

partial association

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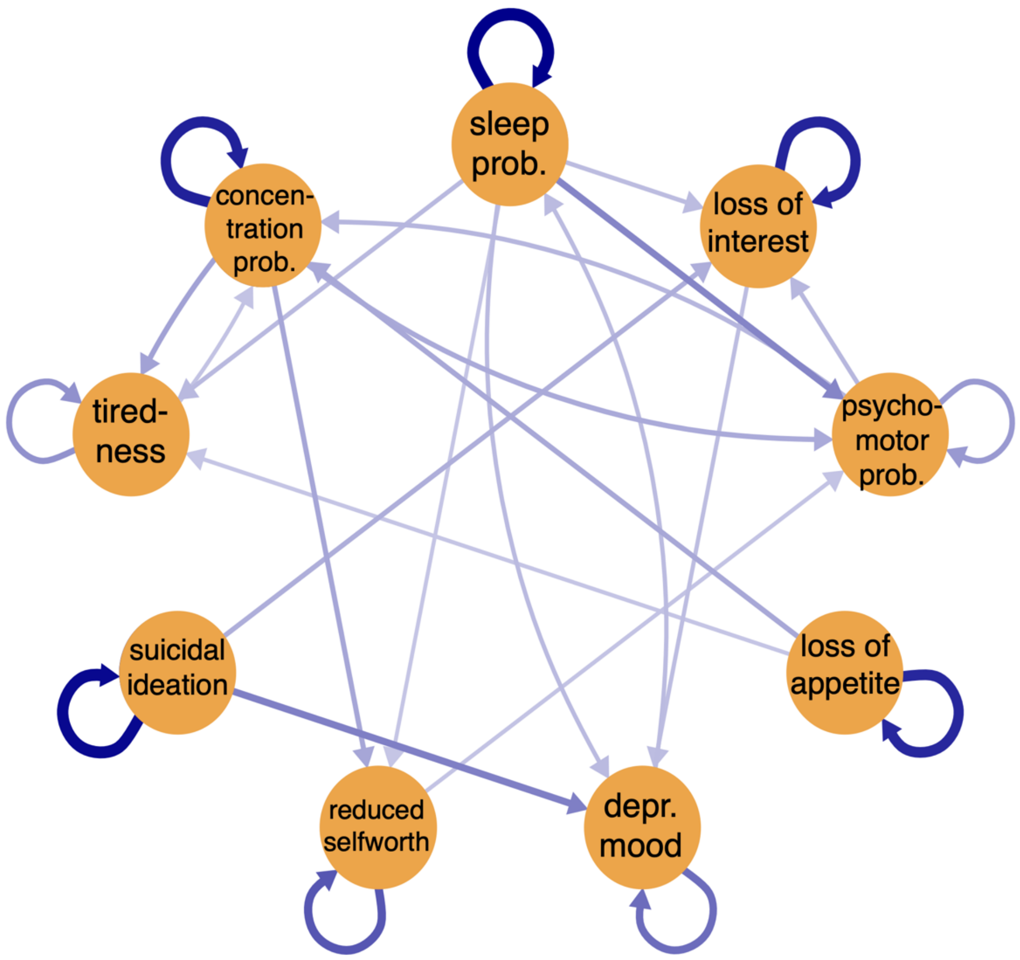
Psychometric network of MDD symptoms



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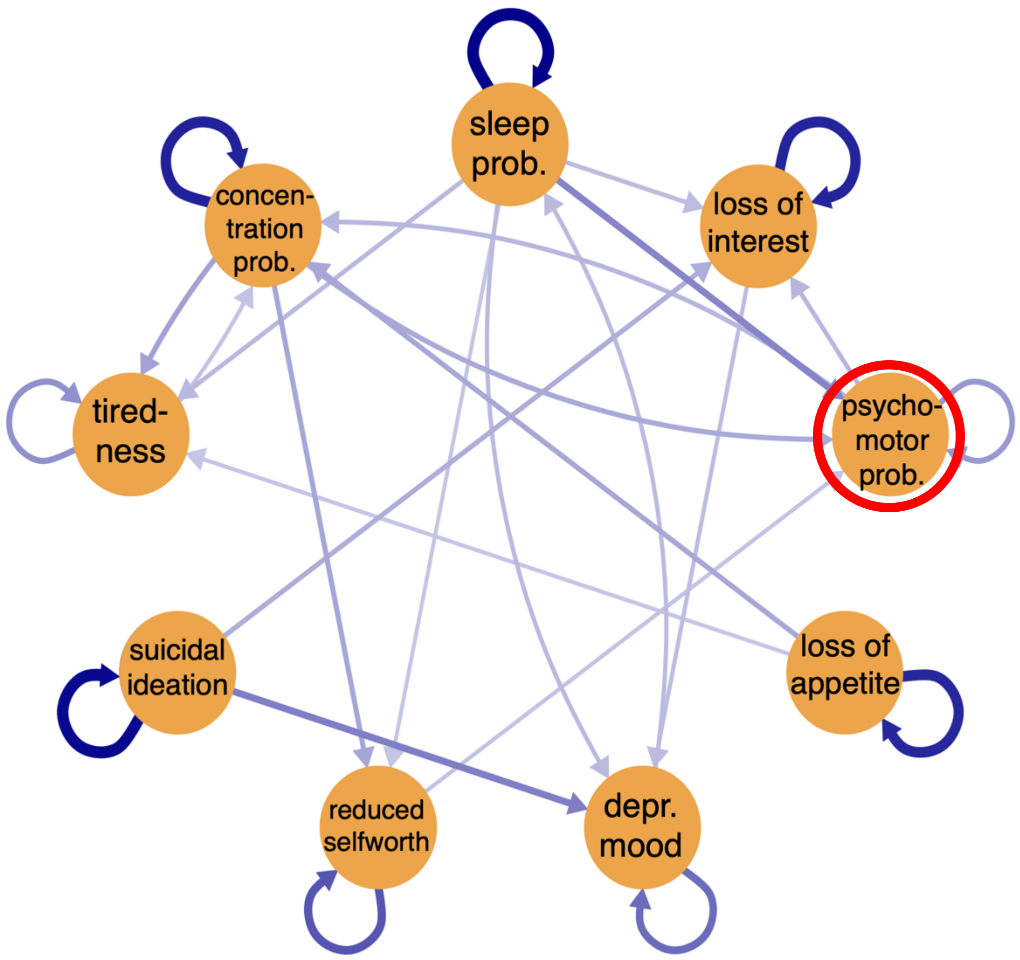

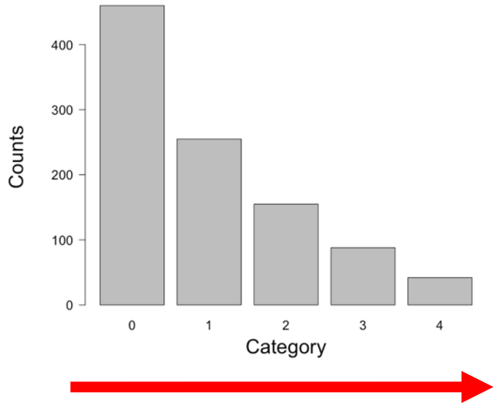

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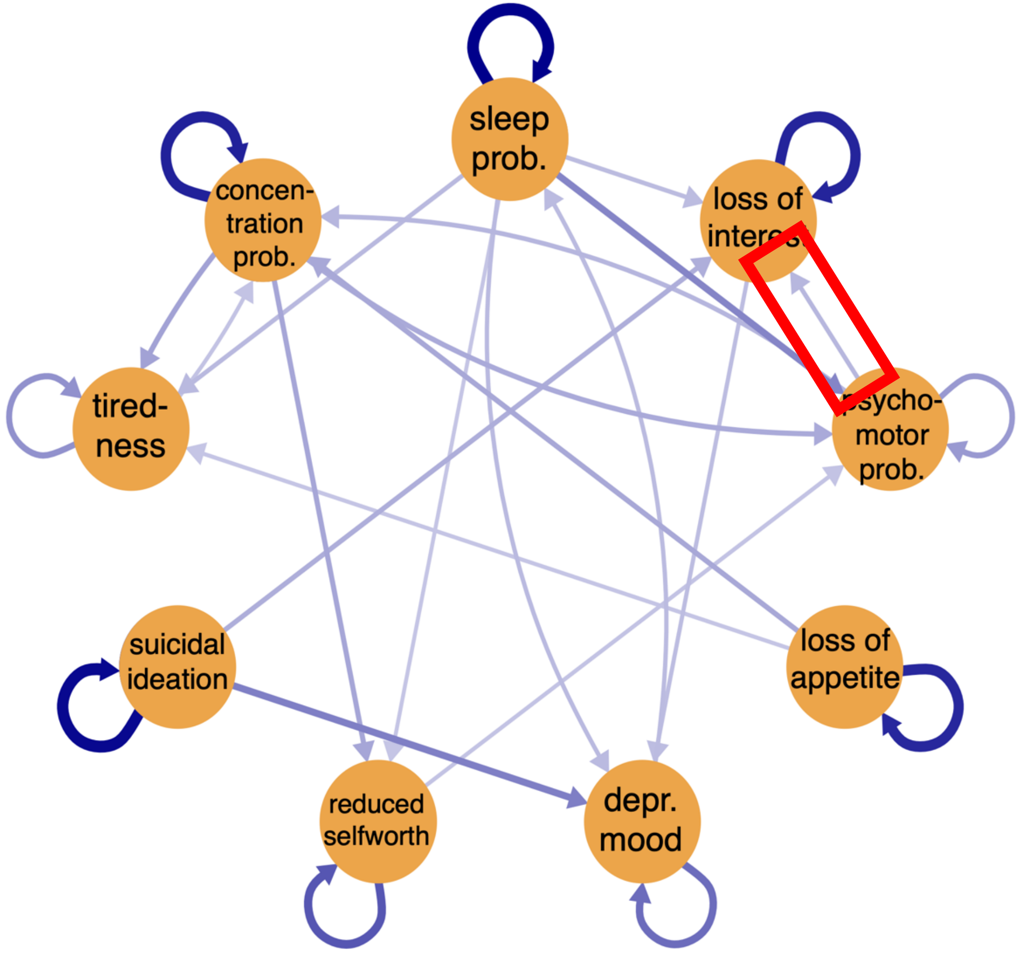

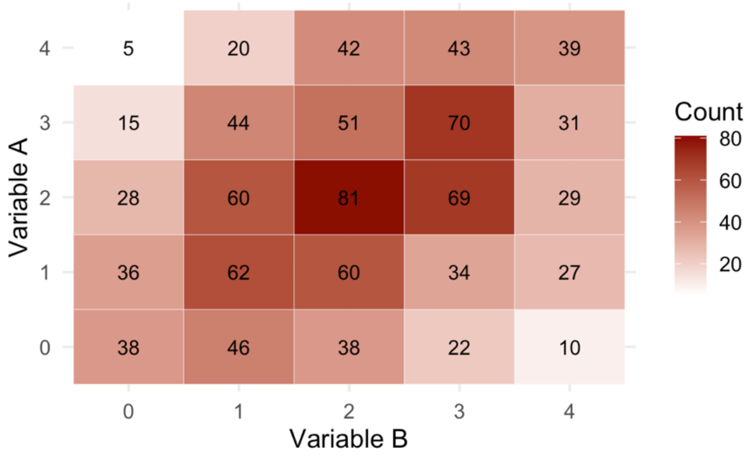

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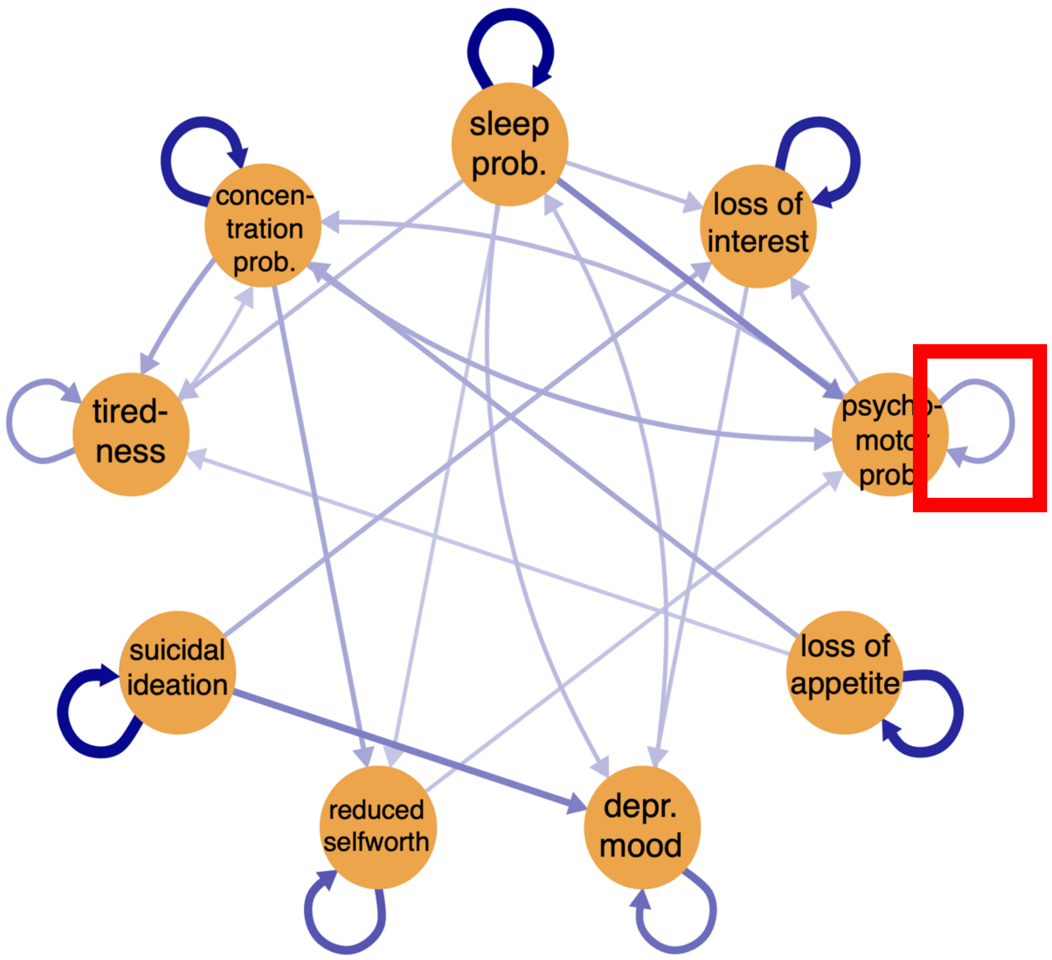

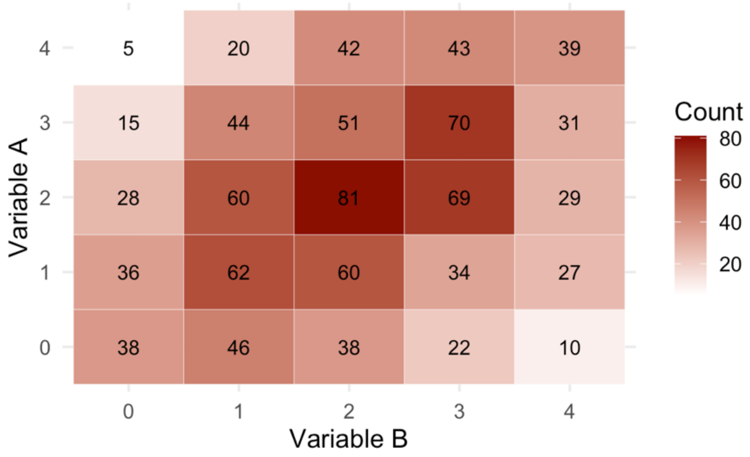

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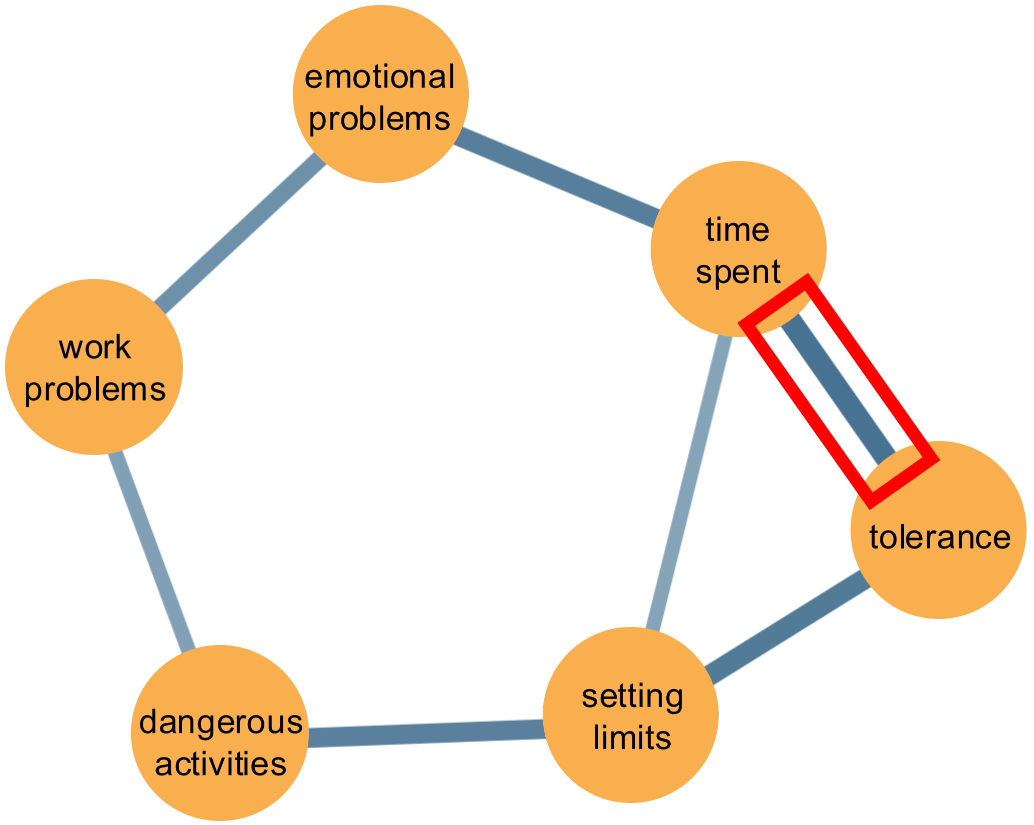
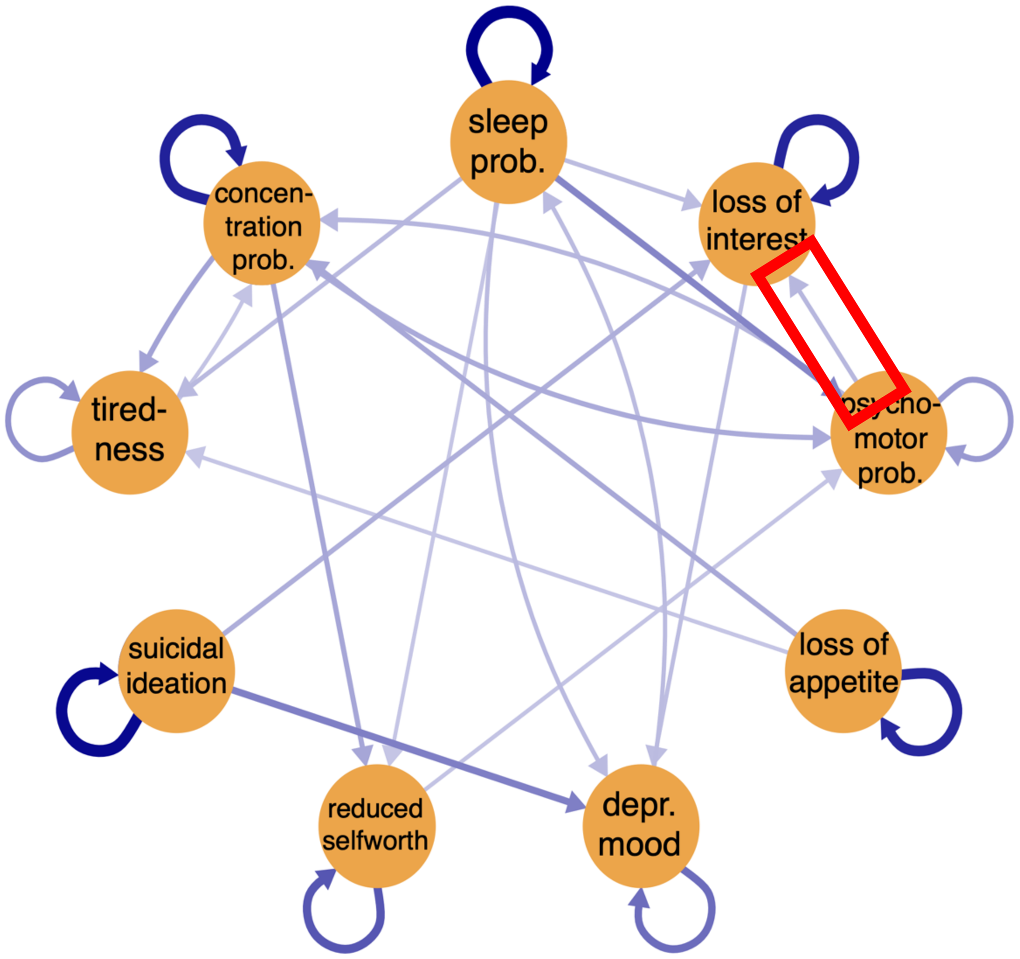
Absent edges give structure to the model

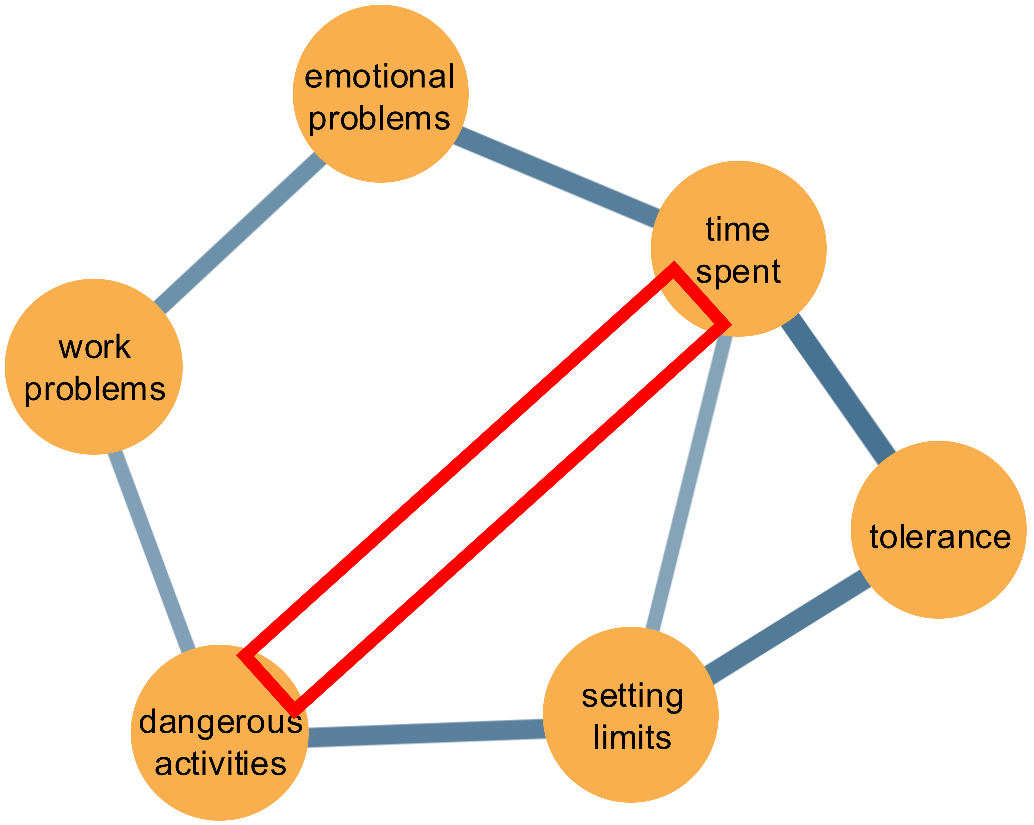


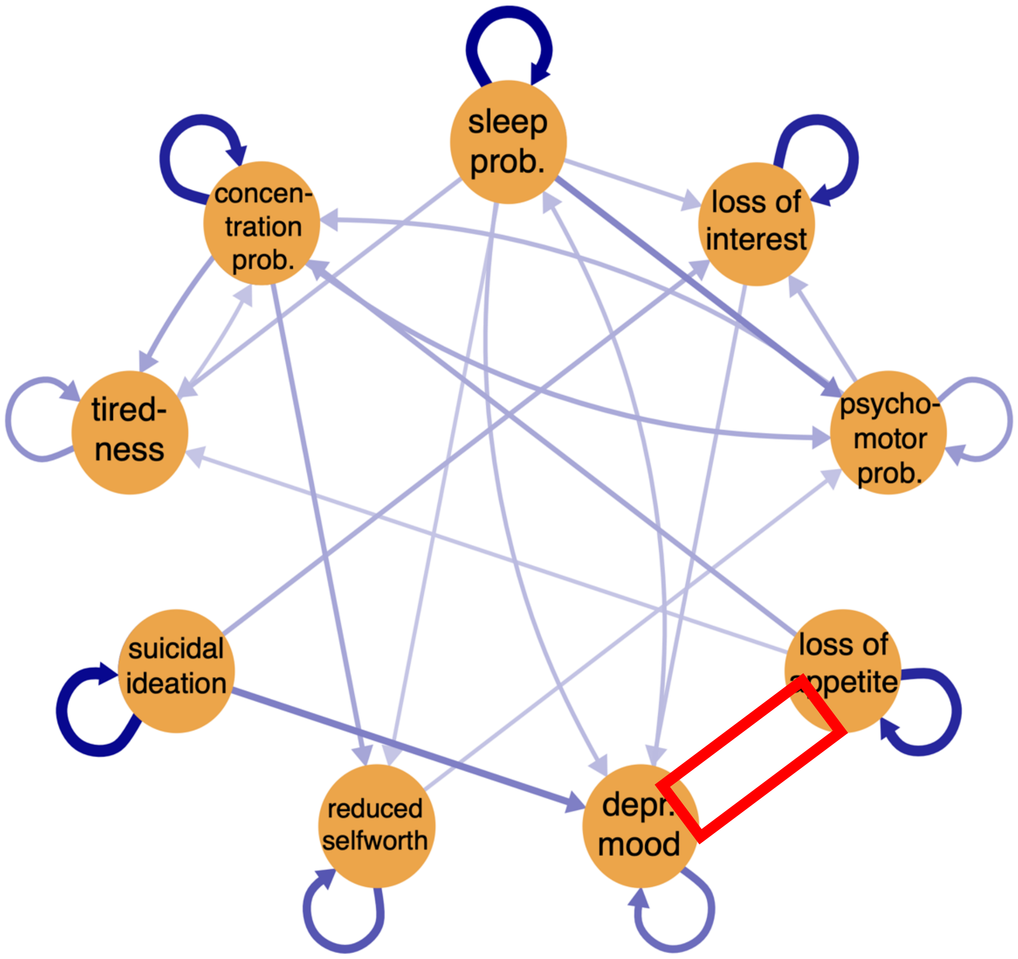
conditional dependence



Granger or predictive causality

Absent edges give structure to the model





conditional independence

no Granger or predictive causality

Network psychometrics

Network modeling aims to:

- Conceptualize psychological constructs as systems of interacting components.
- Represent and analyze these systems using statistical models.
- Interpret systems substantively through theory.
- Connect empirical data with theory in a mutually informative way.

Network analysis aims to:

- Estimate the structure of the network.
- Quantify the strength of relations (parameters).
- Assess the robustness and uncertainty of the network inferences.

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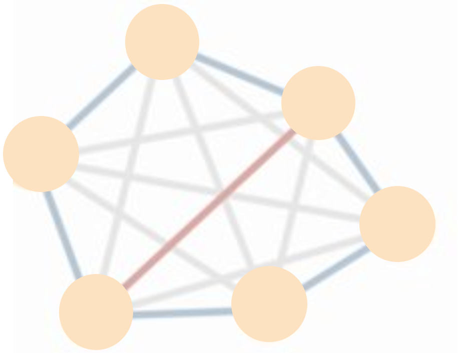

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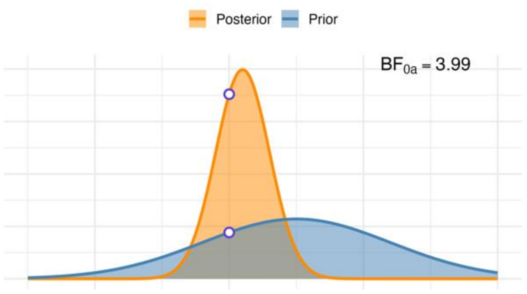


Bayesian Statistics

Bayesian statistics

The **Bayesian approach** requires a statistical model to make predictions of future data.

For a model to make predictions, we need to assign values to its parameters.

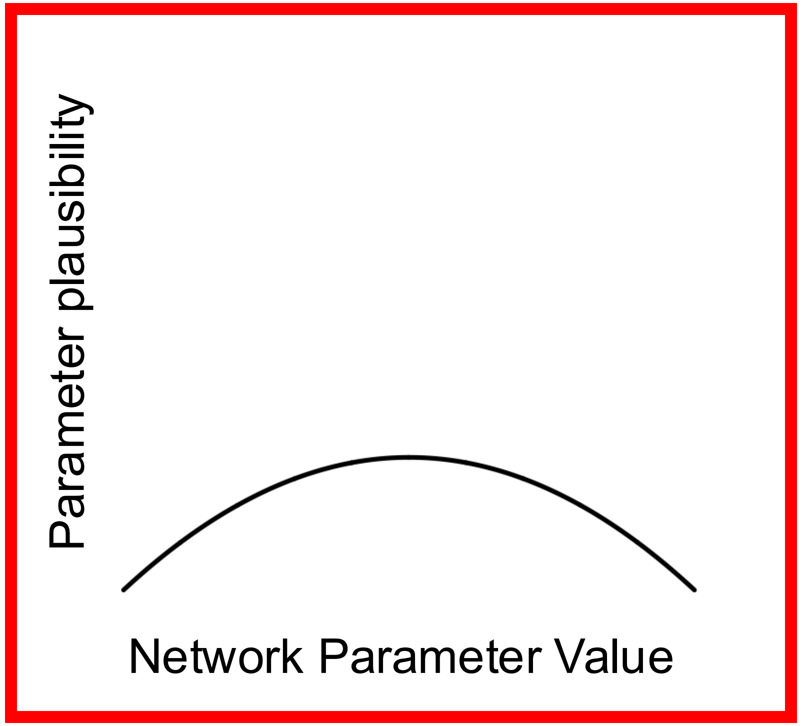
But we often do not know what these values are: they are the thing we want to learn about.

We therefore assign prior distributions to a model’s parameters to reflect the uncertainty about their true value.

Prior plausibility

Predict

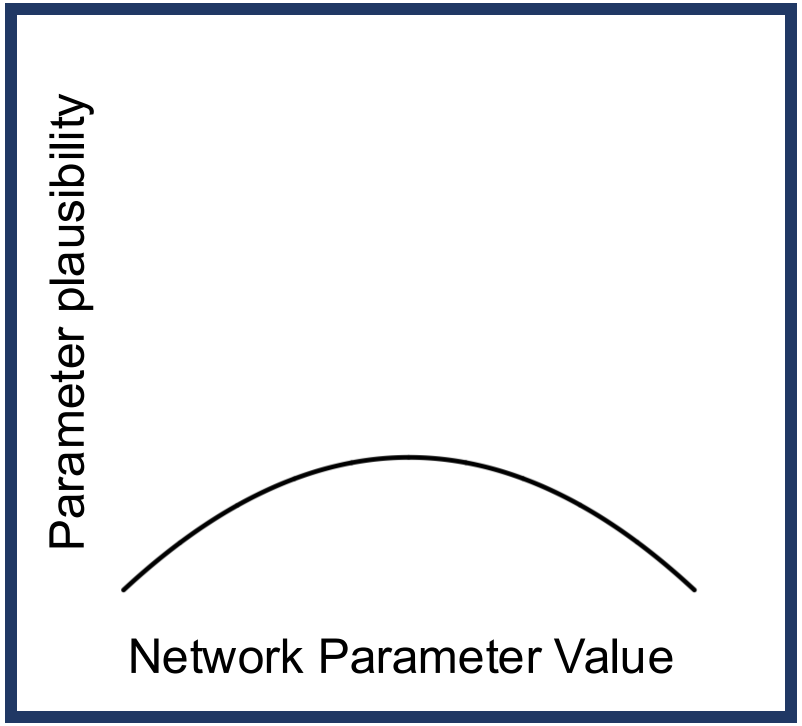
Prior

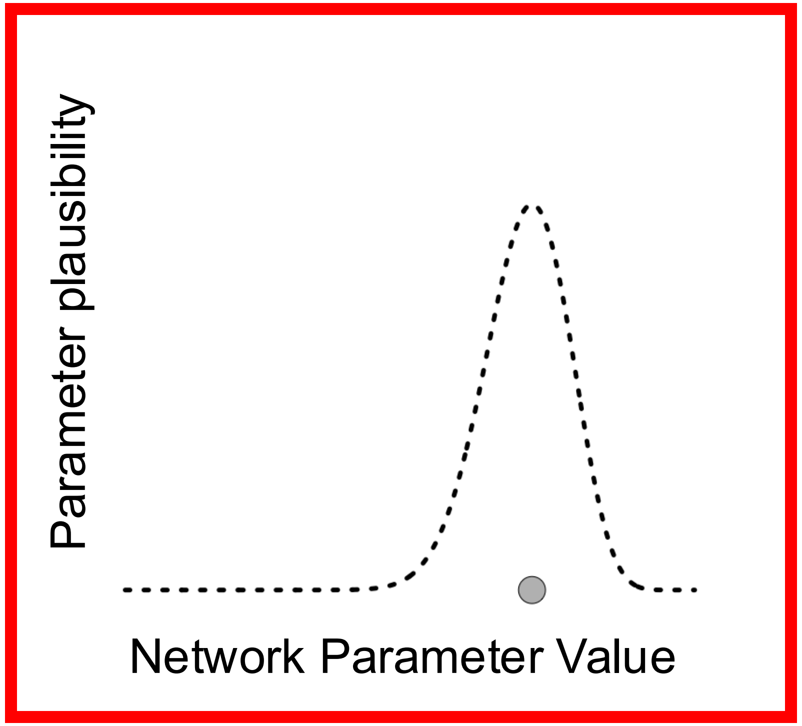


Evaluate predictive success

Predict | Observe

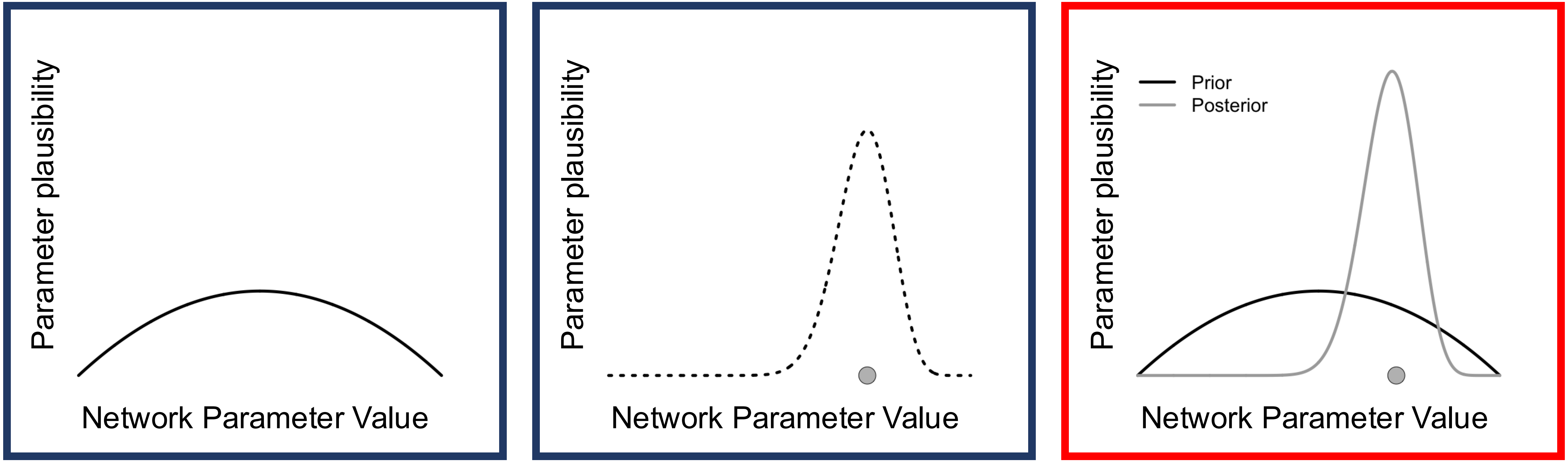
Prior | *Predictive Update Factor*





Posterior plausibility

Predict	Observe	Update
Prior	*Predictive Update Factor*	*Posterior*



Bayesian graphical modeling
(Bayesian network psychometrics)

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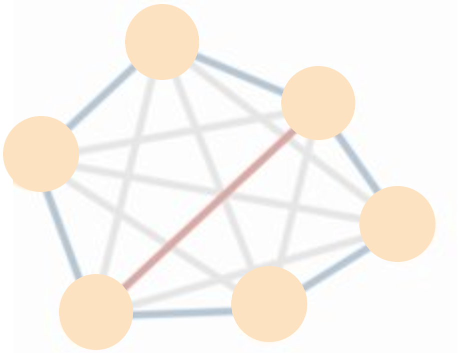
Bayesian network analysis aims to:
- Estimate posterior plausibility of network structures.
- Quantify the strength of relations (parameters) with full distributions.
- Make the uncertainty in results explicit and carry it into any conclusions.
- Develop robust procedures for network estimation and comparison.

Introduction to the workshop



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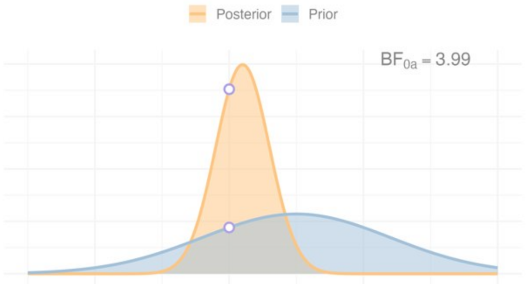


Bayesian Statistics

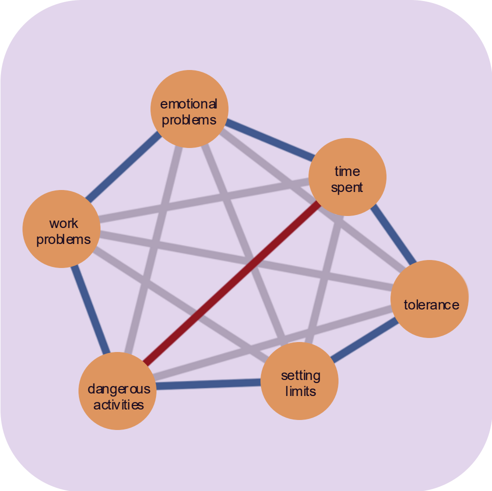
This Workshop

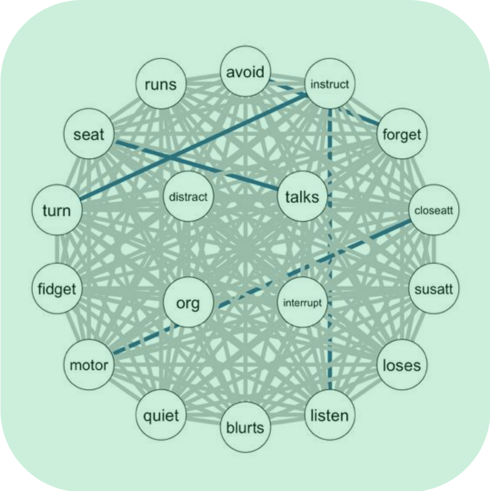
image: Flaticon.com

The workshop

Session I
Monday morning

- estimate posteriors
- test cond. dep.



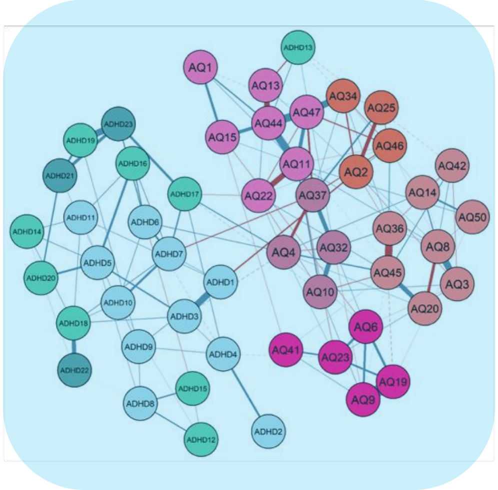


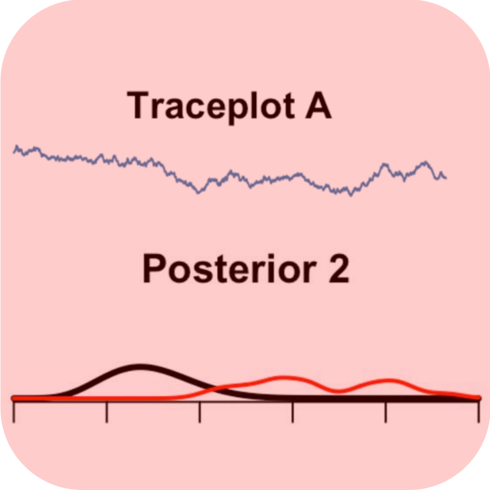
Session III
Tuesday morning

- Savage-Dickey
- group comparison

Session II
Monday afternoon

- topological priors
- clustering





Session IV
Tuesday afternoon

- MCMC diagnostics
- MCMC control pars.

By the end of the workshop, you will be able to:

- **Explain** the (*benefits of the*) Bayesian approach to network analysis.
- **Estimate and interpret** network models with **R** and **JASP**.
- **Understand** how priors affect network structure and complexity.
- **Examine** group differences using edge-specific Bayes factors.
- **Evaluate** convergence and precision using MCMC diagnostics.
- **Apply** the full Bayesian workflow from prior specification and model estimation to inference and interpretation.

Install Software: R

https://cran.r-project.org/

image: Flaticon.com

Install Software: R

```
> install.packages("easybgm")
```

image: Flaticon.com

Install Software: R

```
> install.packages("bgms")
```

image: Flaticon.com

Install Software: JASP (optional)

https://jasp-stats.org/

image: Flaticon.com

Workshop Material

LL

https://github.com/Bayesian-Graphical-Modelling-Lab/BGM_Workshops/tree/5thFSoIM